MATH 6112. FALL 2026. ASSIGNMENT 3.

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Problem 1

(a) We have that the product of two Hermitian matrices may not be Hermitian.

$$\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & 2\end{pmatrix} = \begin{pmatrix}1 & 2\\ 1 & 2\end{pmatrix} \neq \begin{pmatrix}1 & 1\\ 2 & 2\end{pmatrix}$$

(b) Define

$$A = \begin{pmatrix}2 & 0\\ 0 & 10\end{pmatrix}, B = \begin{pmatrix}1 & 2\\ 2 & 5\end{pmatrix}.$$

Trivially $A \succ 0$. We have that $B_{11} = \det(B) = 1 > 0$ and thus, $B \succ 0$. Since $(A-B)_{11} = 1 > 0$ and $\det(A-B) = 1 > 0$, $A \succ B$. However, $(A^2 - B^2)_{11} = -1 < 0$ and hence, $A^2 \not\succ B^2$. This is a valid counterexample.

(c) Define

$$A = \begin{pmatrix}1 & 1\\ 1 & 2\end{pmatrix}, B = \begin{pmatrix}1 & -2\\ -2 & 5\end{pmatrix}.$$

We have that $\det(B) = \det(A) = A_{11} = B_{11} = 1 > 0$ and thus $A \succ 0$, $B \succ 0$. However, $(AB + BA)_{11} = -2 < 0$ and hence, $AB + BA \not\succ 0$. This is a valid counterexample.

(d) Define $B = (1\ 0)^\top$ and $C = (1\ 2)^\top$. We have that

$$CB^\top = \begin{pmatrix}1 & 0\\ 2 & 0\end{pmatrix} =: H, \quad H^\top H = \begin{pmatrix}5 & 0\\ 0 & 0\end{pmatrix} \implies \Sigma(CB^\top) = \left\{\sqrt{5}, 0\right\}, \qquad BC^\top = (1).$$

This gives a valid counterexample.

(e) Define

$$A = \begin{pmatrix}2 & 0\\ 0 & 1\end{pmatrix}, B = \begin{pmatrix}1 & 0\\ 0 & 2\end{pmatrix} \implies A - B = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

Then $A - B$ is clearly not positive semidefinite.

Problem 2

Since A, B are Hermitian, there exists unitary matrices U, V such that

$$A = U^*\Lambda_A U \qquad \text{and} \qquad B = V^*\Lambda_B V,$$

where $\Lambda_A = \operatorname{diag}\{\lambda_i(A)\}_{i\in[n]}$ and $\Lambda_B = \operatorname{diag}\{\lambda_i(B)\}_{i\in[n]}$. Therefore, we have that

$$V^*UAU^*V - B = V^*(UAU^* - \Lambda_B)V = V^*(\Lambda_A - \Lambda_B)V.$$

Let $W = U^*V$ and then W is also unitary. Since $\lambda_i(A) \geq \lambda_i(B)$ for all $i \in [n]$, $\Lambda_A - \Lambda_B \succeq 0$ and hence, $W^*AW \succeq B$.

Problem 3

By the singular value decomposition, there exists unitary matrices U, V such that $A = U\Sigma V$, where $\Sigma = \operatorname{diag}\{\sigma_i\}_{i\in[n]}$. Let $W = (w_{ij})_{i,j\in[n]} = VU$. Then W is unitary. Therefore, we have that

$$|\operatorname{tr}(A)| = |\operatorname{tr}(U\Sigma V)| = |\operatorname{tr}(W\Sigma)| \le \sum_{i=1}^{n} \sigma_i |w_{ii}| \le \sum_{i=1}^{n} \sigma_i,$$

where the last inequality follows from $|w_{ii}| \le 1$ since W is unitary.

Problem 4

Let n be the size of the matrix. Observe that

$$M = \begin{pmatrix} A^{1/2} & 0 \\ 0 & B^{1/2} \end{pmatrix} \begin{pmatrix} I_n & I_n \\ I_n & I_n \end{pmatrix} \begin{pmatrix} A^{1/2} & 0 \\ 0 & B^{1/2} \end{pmatrix} \implies \operatorname{rank}(M) \le \operatorname{rank} \begin{pmatrix} I_n & I_n \\ I_n & I_n \end{pmatrix} = n.$$

Therefore, $\operatorname{rank}(M) < 2n$ and M is not full rank, implying $\det(M) = 0$. Also, for any $x_1, x_2 \in \mathbb{C}^n$, we have that

$$\begin{pmatrix} x_1^* & x_2^* \end{pmatrix} \begin{pmatrix} A & A^{1/2}B^{1/2} \\ B^{1/2}A^{1/2} & B \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \left\| A^{1/2}x_1 + B^{1/2}x_2 \right\|_2^2 \ge 0.$$

Hence, $M \succeq 0$.

WLOG, assume A is positive definite. Then $A^{1/2}$ is also positive definite. Direct calculation gives

$$\forall x_2 \in \mathbb{C}^n, \quad \begin{pmatrix} A & A^{1/2}B^{1/2} \\ B^{1/2}A^{1/2} & B \end{pmatrix} \begin{pmatrix} -A^{-1/2}B^{1/2}x_2 \\ x_2 \end{pmatrix} = 0$$

$$\begin{pmatrix} A & A^{1/2}B^{1/2} \\ B^{1/2}A^{1/2} & B \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \implies A^{1/2}(A^{1/2}x_1 + B^{1/2}x_2) = 0 \implies x_1 = -A^{-1/2}B^{1/2}x_2.$$

Therefore, $\ker(M) = \{(x_1^* \ x_2^*)^* \mid x_1 = -A^{-1/2}B^{1/2}x_2\}$ and the null space of M has dimension n.

Problem 5

Let $M(A, B)$ be the matrix defined in Problem **(4)**. We have that $M(A, B) \succeq 0$ ("at least one of them be positive definite" is condition is redundant for $M(A, B) \succeq 0$). Define $M'(\varepsilon) = M(A, B) + M(B, A) + \varepsilon I_{2n}$ and thus, $M' \succ 0$ for $\varepsilon > 0$. We write $C = A + B + \varepsilon I \succ 0$ and $D = A^{1/2}B^{1/2} + B^{1/2}A^{1/2}$. Define $S = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix}$, where $X = -C^{-1}D$. Therefore, we have that by definition of X,

$$S^*M'(\varepsilon)S = \begin{pmatrix} C & CX + D \\ X^*C + D & C + X^*CX + X^*D + DX \end{pmatrix} = \begin{pmatrix} C & 0 \\ 0 & C - DC^{-1}D \end{pmatrix}.$$

Since the congruent transformation preserves the positive definiteness, we have that $S^*M'(\varepsilon)S \succ 0$ and thus, $C - DC^{-1}D \succ 0$. By congruent transformation,

$$C - DC^{-1}D \succ 0 \implies I \succ \left(C^{-1/2}DC^{-1/2}\right)^2.$$

Let $H = C^{-1/2}DC^{-1/2}$ and then $H = H^*$. Therefore, $H^2 = HH^* \succeq 0$ and we have that

$$I \succ H^2 \succeq 0 \implies 1 \geq \det(H)^2 \iff \det(C)^2 \geq \det(D)^2.$$

Since $C \succ 0$ and hence $\det(C) > 0$, we have that $\det(C) \geq \det(D)$. Letting ε tend to 0, we conclude that

$$\det(A + B) \geq \det\left(A^{1/2}B^{1/2} + B^{1/2}A^{1/2}\right).$$

Problem 6

For any $x \in \mathbb{R}^n$, we have that

$$x^\top (A \circ B)x = \sum_{i=1}^{n}\sum_{j=1}^{n} x_i a_{ij} b_{ij} x_j = \mathrm{Tr}\left(D_x A D_x B^\top\right) = \mathrm{Tr}\left(B^{1/2}D_x A^{1/2}A^{1/2}D_x B^{1/2}\right).$$

Let $M = (m_{ij})_{i,j\in[n]} = B^{1/2}D_x A^{1/2}$. Then $M^\top = A^{1/2}D_x B^{1/2}$. Therefore,

$$x^\top (A \circ B)x = \mathrm{Tr}\left(MM^\top\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} m_{ij}^2 \geq 0,$$

which implies that $A \circ B$ is positive semidefinite.

Furthermore, if $A \succ 0$ and $B \succ 0$, then both $A^{1/2}$ and $B^{1/2}$ are full rank. If $x \neq \mathbf{0}$, then $\mathrm{rank}(D_x) \geq 1$ and therefore,

$$\mathrm{rank}(M) = \mathrm{rank}(B^{1/2}D_x A^{1/2}) = \mathrm{rank}(D_x) \geq 1,$$

implying that $M \neq 0$ and thus, $\mathrm{Tr}\left(MM^\top\right) \neq 0$. In this case, $x^\top (A \circ B)x > 0$ and $A \circ B \succ 0$.

Acknowledgement

I used ChatGPT to help construct and check the counterexamples in Problems **1(b)** and **1(c)**.